

Installation & Set up Guide

Bookstagram

"Why just read when you can create!"

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1. Prerequisites

Before installing Bookstagram, make sure the following software is installed on your machine:

- Node.js v18 or higher - <https://nodejs.org>
- npm v9 or higher (comes with Node.js)
- MongoDB Atlas account - <https://mongodb.com/atlas>
- Git - <https://git-scm.com>
- A code editor such as VS Code

2. Clone the Repository

Open your terminal and run the following command to clone the project:

`git clone https://github.com/KhushiBhatt22/W26_4495_S2_KhushiB.git`

3. Backend Setup

3.1 Navigate to the Backend Folder

- **`cd Implementation/backend`**

3.2 Install Dependencies

- Run the following command to install all required backend packages:
- **`npm install`**

3.3 Create the Environment File

There is an empty .env file where you can see variables such as:

Variable	Example Value	Description
PORT	8000	Port the backend server runs on
MONGO_URI	mongodb+srv://...	Your MongoDB Atlas connection string
JWT_SECRET	your_secret_key	Secret key for JWT token signing
GEMINI_API_KEY	AIza...	Google Gemini API key for AI writing
HUGGING_FACE_API_KEY	hf_...	Hugging Face API key for image generation
CLOUDINARY_CLOUD_NAME	your_cloud	Cloudinary cloud name for image storage

<code>CLOUDINARY_API_KEY</code>	<code>123456789</code>	Cloudinary API key
<code>CLOUDINARY_API_SECRET</code>	<code>abc123...</code>	Cloudinary API secret

Never commit your .env file to GitHub. Make sure it is listed in your .gitignore file.

3.4 Start the Backend Server

- Run the following command to start the backend in development mode:
- **`npm run dev`**
- The backend server will start on: <http://localhost:8000>

4. Frontend Setup

4.1 Navigate to the Frontend Folder

- Open a new terminal window and run:
- **`cd Implementation/frontend/bookstagram`**

4.2 Install Dependencies

- **`npm install`**

4.3 Start the Frontend

- **`npm run dev`**
- The frontend will start on: <http://localhost:5173>
- Open your browser and navigate to <http://localhost:5173> to see the application.

5. Database Setup (MongoDB Atlas)

1. Go to <https://mongodb.com/atlas> and create a free account.
2. Create a new Cluster (free tier M0 is sufficient).
3. Go to Database Access → Add a database user with username and password.
4. Go to Network Access → Add IP Address → Allow Access from Anywhere (0.0.0.0/0).
5. Click Connect → Drivers → Copy the connection string and paste it as MONGO_URI in your .env file.

Replace <password> in the connection string with your actual database user password.

6. JWT Secret Key

Generate a secure random JWT secret key using this command in your terminal:

```
node -e "console.log(require('crypto').randomBytes(64).toString('hex'))"
```

Copy the output and paste it as the value of JWT_SECRET in your .env file.

7. API Keys Setup

7.1 Google Gemini API Key

6. Go to <https://aistudio.google.com>
7. Sign in with your Google account.
8. Click Get API Key → Create API Key → Copy and paste into .env as GEMINI_API_KEY.

7.2 Hugging Face API Key

9. Go to <https://huggingface.co> and create a free account.
10. Go to Settings → Access Tokens → New Token → Copy and paste as HUGGING_FACE_API_KEY.

7.3 Cloudinary

11. Go to <https://cloudinary.com> and create a free account.
12. From your Dashboard, copy Cloud Name, API Key, and API Secret into your .env file.

8. Running the Full Application

You need TWO terminal windows running simultaneously:

Terminal 1 - Backend:

```
cd Implementation/backend
```

```
npm run dev
```

Terminal 2 - Frontend:

```
cd Implementation/frontend/bookstagram
```

```
npm run dev
```

Then open your browser and go to: <http://localhost:5173>

9. Technology Stack

Frontend	Backend	Services
• React.js (Vite)	• Node.js + Express	• Google Gemini AI

Bookstagram

• React Router DOM • Tailwind CSS • Axios • Lucide React Icons	• MongoDB + Mongoose • JWT Authentication • Multer (file upload) • bcryptjs	• Hugging Face (FLUX) • Cloudinary • MongoDB Atlas
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